



EMPLOYEE PAYROLL SYSTEM

Presented by Group D

Evening



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INTRODUCTION

The Employee Payroll System is a menu-driven C program developed to automate employee salary processing.

The system helps to:

- Register new employees
 - Calculate salaries
 - Apply overtime payments
 - Apply deductions
 - Generate payroll payments
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PROBLEM STATEMENT AND OBJECTIVES

Many payroll
systems are:

- Slow
- Error-prone
- Difficult to manage

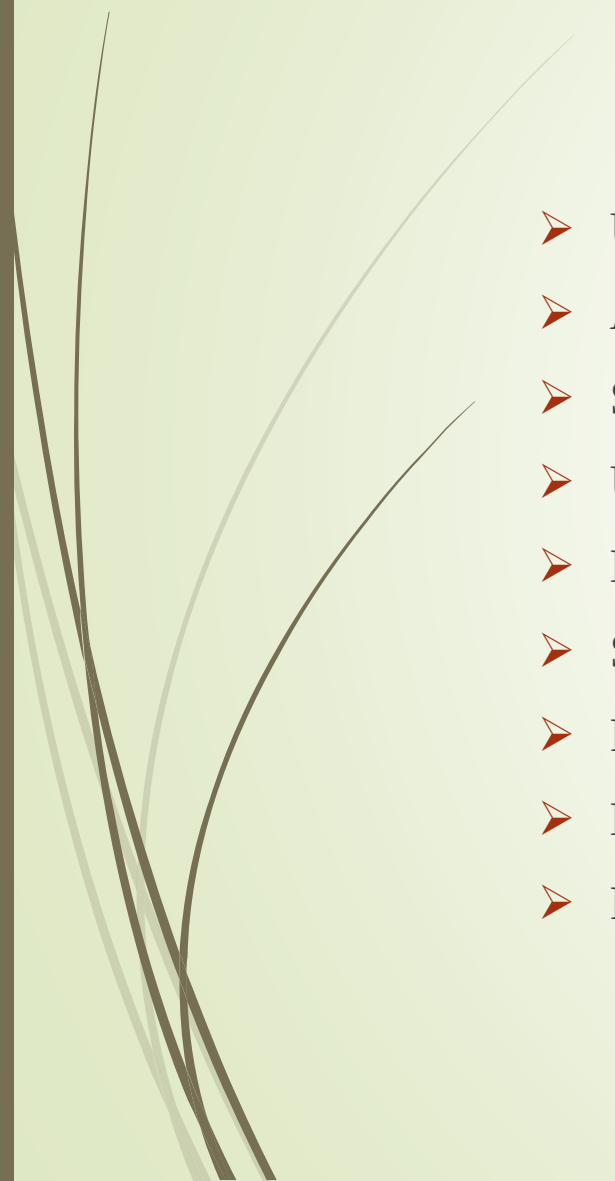
An automated payroll system is
needed to improve efficiency
and accuracy

Objectives

- Register employee records
- Calculate salaries
- Apply overtime deductions
- Generate payroll summaries
- Store employee records permanently.



SYSTEM FEATURES

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- User authentication/login
 - Add employee records
 - Search employee records
 - Update employee details
 - Delete employee records
 - Salary calculation
 - Payroll summary generation
 - Highest-paid employee identification
 - File handling for permanent storage

TECHNOLOGIES AND CONCEPTS USED

➡ C programming language

CONCEPTS APPLIED:

➤ Variables

➤ Loops

➤ Functions

➤ Arrays

➤ Structures

➤ Conditional statements

➤ File handling

➤ validation



EMPLOYEE STRUCTURE AND SAMPLE CALCULATION LOGIC

Employee structure:

```
struct Employee{  
    int id;  
    char name[50];  
    char position[30];  
    float hourlyRate;  
    float normalHoursLimit;  
    float hoursWorked;  
  
} Employee;
```

Sample salary calculation logic:

Formulas used:

Overtime hours = $\max(0, \text{hoursWorked} - \text{normalHoursLimit})$

Grosspay = $(\text{normalHoursLimit} * \text{rate}) + (\text{overtimeHours} * \text{rate} * 1.5)$

Deductions

≤ 2000 – 10%

2001-4000 – 15%

4000 – 20%

Netpay = gross-deductions



CHALLENGES AND SOLUTIONS

CHALLENGES

- Buffer issues after
- Duplicate employee IDs
- User desires to keep old value on update
- Excessive hours detection

SOLUTIONS

- Function `clearInputBuffer()`
- `isIdUnique()` checks all existing IDs
- Check if first char is `'\n'`
- Absolute (200) + relative(>1.5* limit)



CONCLUSION

The Employee Payroll system was successfully developed using the C programming language.

Achievements

- Automated payroll processing
- Accurate salary calculations
- Reduced manual errors
- Improved employee management
- Demonstrated major programming concepts



THANK YOU FOR LISTENING